The documentation and process conversion measures necessary to comply with this revision shall be completed by 21 February 1994

INCH-POUND

MIL-S-19500/518B(USAF)
21 November 1994
SUPERSEDING
MIL-S-19500/518A(USAF)
14 May 1984

MILITARY SPECIFICATION
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER
TYPE 2N3766, 2N3767 JAN, JANTX, AND JANTXV

This specification is approved for use by the Rome Laboratory, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for NPN silicon, power transistors. Three levels of product assurance are provided for each device type as specified in MIL-S-19500.

1.2 Physical dimensions. See 3.3 (TO-66).

1.3 Maximum ratings.

Type	P_T T_C = +25°C 1/	V_{CBO}	V_{CEO}	V_{EBO}	I_B	I_C	T_{OP} and T_{stg}	$R_{\theta JC}$
	W	V dc	V dc	V dc	A dc	A dc	°C	°C/W
2N3766	25	80	60	6	2	4	-55 to +200	7
2N3767	25	100	80	6	2	4	-55 to +200	7

1/ Between T_C = +25°C and T_C = +200°C, linear derating factor (average) = 143 m/W°C.

1.4 Primary electrical characteristics.

Type	h_{FE2} 1/		h_{FE3} 1/		C_{obo}		$\|h_{fe}\|$		$V_{BE(sat)}$	
	V_{CE} = 10 V dc I_C = 1 A dc		V_{CE} = 5 V dc I_C = 500 mA dc		V_{CB} = 10 V dc I_E = 0 0.1 MHz ≤ f ≤ 1 MHz		V_{CE} = 10 V dc I_C = 500 mA dc f = 10 MHz		I_C = 1 A dc I_B = 0.1 A dc	
	Min	Max	Min	Max	Min pF	Max pF	Min	Max	Min V dc	Max V dc
2N3766	20		40	160		50	1	8		1.5
2N3767	20		40	160		50	1	8		1.5

Type	$V_{CE(sat)1}$		$V_{CE(sat)2}$		Switching			
	I_C = 1 A dc I_B = 0.1 A dc		I_C = 500 mA dc I_B = 50 mA dc		t_{on}		t_{off}	
					(see table I and figure 1 herein)			
	Min	Max	Min	Max	Min	Max	Min	Max
					µs	µs	µs	µs
2N3766		2.5		1.0		.25		2.5
2N3767		2.5		1.0		.25		2.5

1/ Pulsed (see 4.5.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ELDT, 1507 Wilmington Pike, Dayton, OH 45444-5765 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

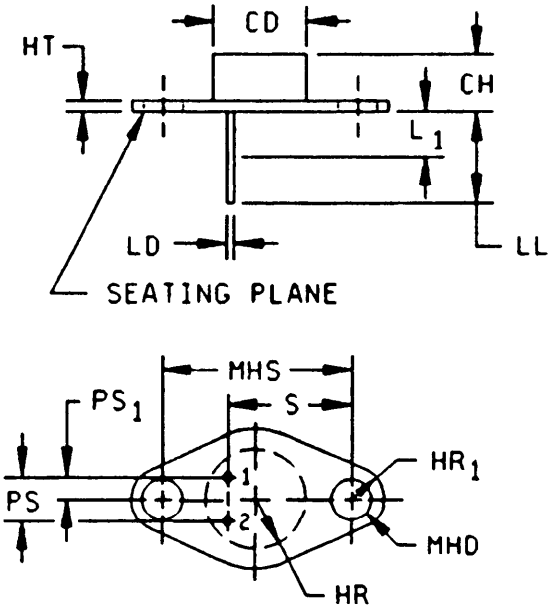
AMSC N/A

FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T02:42Z
Check the source to verify that this is the current version before use.

Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
CD		.620		15.76	
CH	.250	.340	6.35	8.64	
HR		.350		8.89	4
HT	.050	.075	1.27	1.91	
HR_1	.115	.145	2.92	3.68	4
LD	.028	.034	0.71	0.86	4,6
LL	.360	.500	9.14	12.70	
L_1		.050		1.27	6
MHD	.142	.152	3.61	3.86	4
MHS	.958	.962	24.33	24.43	
PS	.190	.210	4.83	5.33	3
PS1	.093	.107	2.36	2.72	3
S	.570	.590	14.48	14.99	



NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) +.005 inch (0.13 mm) -.000 inch (0.00 mm) below seating plane. When gauge is not used, measurement will be made at the seating plane.
4. Two places.
5. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
6. Lead diameter shall not exceed twice LD within L_1.
7. Lead number 1 is the emitter, lead 2 is the base, case is the collector.
8. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions (similar to TO-66).

Source: https://assist.dla.mil -- Downloaded: 20261O-19T02:42Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figure 1 herein (TO-66).

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500, MIL-STD-750 and herein. When a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:42Z
Check the source to verify that this is the current version before use.

4.3 Screening (JANTX and JANTXV levels). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement JANTX and JANTXV levels only
11	I_{CEX1} and h_{FE2};
12	See 4.3.1
13	Subgroup 2 of table I herein. ΔI_{CEX1} = 100 percent of initial value or 1 μA dc, whichever is greater; Δh_{FE} = 25 percent of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

T_J = +187.5°C ±12.5°C; V_{CB} ≥ 10 V dc; T_A ≤ +100°C.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with group A (table I), subgroup 2 herein.

Subgroup	Method	Condition
B3	1027	V_{CB} ≥ 10 V dc; ΔT_J between cycles ≥ +100°C; $t_{on} = t_{off}$ = 3 minutes for 2,000 cycles.
B5	3131	See 4.5.3.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500. Electrical measurements (end-points) shall be in accordance with group A (table I), subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Test condition A, weight = 10 pounds, time = 15 s.
C6	1027	V_{CB} ≥ 10 V dc; ΔT_J between cycles ≥ +100°C; $t_{on} = t_{off}$ = 3 minutes for 6,000 cycles. No heat sink or forced-air cooling on device shall be permitted.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Coil selection for safe operating area (SOA) tests. In selecting coils for use in clamped and unclamped inductive SOA tests, prime consideration should be given to the commercially available coil. However, due to the extreme critical nature of the coil in these circuits and wide tolerance of some commercially available coils (+100 percent, -50 percent), it shall be the semiconductor manufacturer's responsibility to prove upon request compliance or equivalency of any coil used (commercial or in-plant designed) to be within (+20 percent, -10 percent) of the specified inductance at the rated current and dc resistance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:42Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Collector to emitter breakdown voltage	3011	Bias condition D; I_C = 100 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N3766				60		V dc
2N3767				80		V dc
Collector to emitter cutoff current	3041	Bias condition D;	I_{CEO}		500	µA dc
2N3766		V_{CE} = 60 V dc				
2N3767		V_{CE} = 80 V dc				
Emitter-base cutoff current	3061	Bias condition D; V_{EB} = 6 V dc	I_{EBO}		500	µA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc;	I_{CEX1}			
2N3766		V_{CE} = 80 V dc			10	µA dc
2N3767		V_{CE} = 100 V dc			10	µA dc
Collector to base cutoff current	3036	Bias condition D;	I_{CBO}			
2N3766		V_{CB} = 80 V dc			10	µA dc
2N3767		V_{CB} = 100 V dc			10	µA dc
Base to emitter non-saturated voltage	3066	Test condition B; V_{CE} = 10 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	V_{BE}		1.5	V dc
Collector to emitter saturated voltage	3071	I_C = 1.0 A dc; I_B = 0.1 A dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		2.5	V dc
Collector to emitter saturated voltage	3071	I_C = 0.5 A dc; I_B = 0.05 A dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		1.0	V dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 50 mA dc; pulsed (see 4.5.1)	h_{FE1}	30		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE3}	40	160	

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:42Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	h_{FE3}	20		
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 1.5 V dc	I_{CEX2}			
2N3766		V_{CE} = 50 V dc			1.0	mA dc
2N3767		V_{CE} = 70 V dc			1.0	mA dc
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 0.5 A dc; pulsed (see 4.5.1)	h_{FE4}	13		
Subgroup 4						
Pulse response:	3251	Test condition A except test circuit and pulse requirements in accordance with figure 2 herein.				
Turn-on time		V_{CC} = 30 V dc; I_C = 0.5 A dc; I_{B1} = 0.05 A dc	t_{on}		.25	us
Turn-off time		V_{CC} = 30 V dc; I_C = 0.5 A dc; I_{B1} = I_{B2} = .05 A dc	t_{off}		2.5	us
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 500 mA dc; f = 10 MHz	$\|h_{fe}\|$	1	8	
Open capacitance (open circuit)	3236	V_{CE} = 10 V dc; I_E = 0; 0.1 MHz ≤ f ≤ 1.0 MHz	C_{obo}		50	pF

See footnote at end of table.

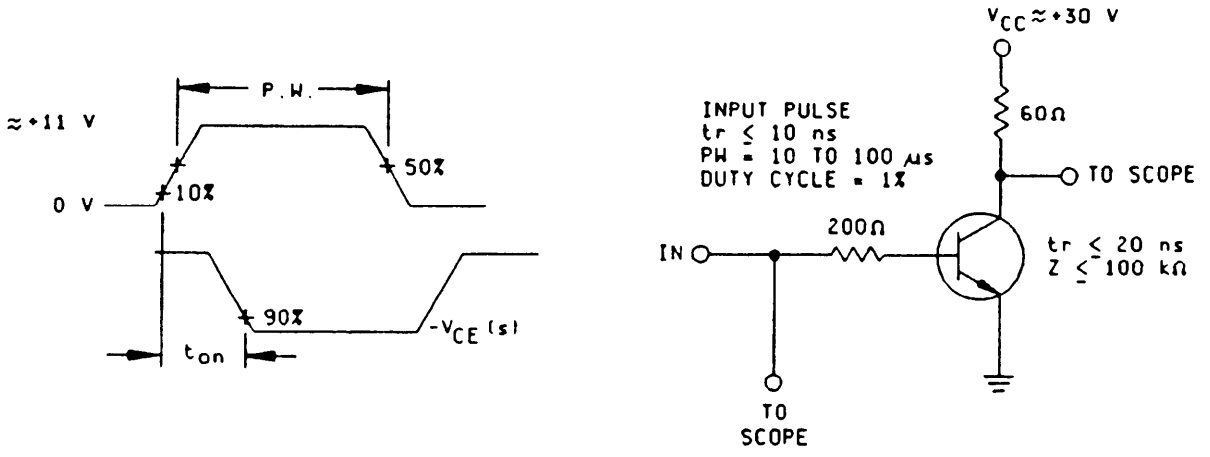
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:42Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

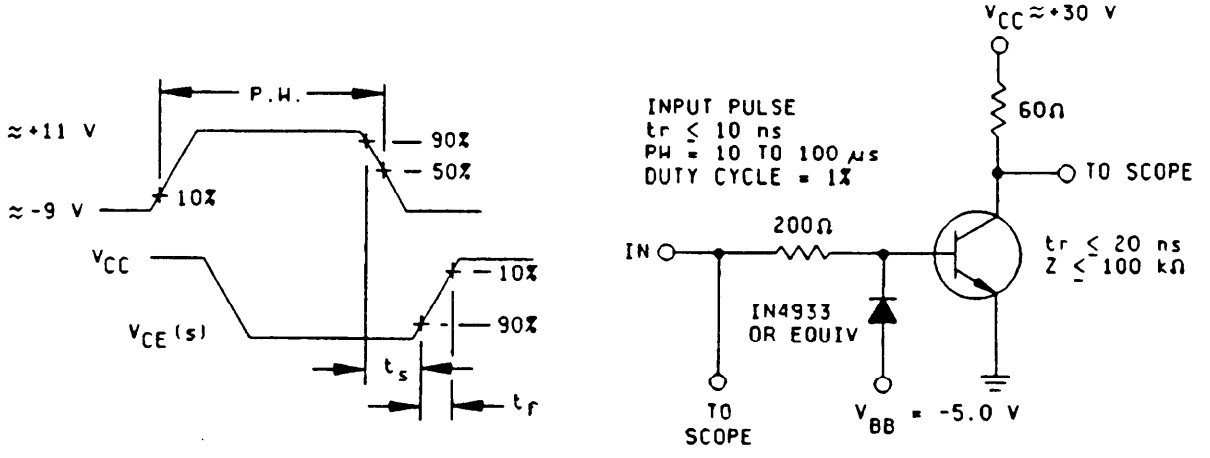
Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
SOA (continuous dc)	3051	T_C = +25°C; power application t = 1 s; 1 cycle; (see figure 3)				
Test 1		V_{CE} = 6.25 V dc; I_C = 4 A dc				
Test 2		V_{CE} = 20 V dc; I_C = 1.25 A dc				
Test 3						
2N3766		V_{CE} = 50 V dc; I_C = 150 mA dc				
2N3767		V_{CE} = 65 V dc; I_C = 150 mA dc				
SOA (clamped switching)	3053	Load condition B; T_A = +25°C; V_{CC} = 20 V dc, t_p = 1.5 ms, R_{BB1} = 5 Ω; (Vary to obtain I_C); duty cycle ≤ 5 percent; I_C = 4 A dc; R_{BB2} = 100 Ω; V_{BB2} = 1.5 V dc; V_{BB1} = 10 V dc; L = 5 mH, at 4A, R of inductor = .5 Ω, and R_{LOAD} = R of inductor. 2/				
2N3766		Clamp voltage = 60 V dc				
2N3767		Clamp voltage = 80 V dc				
		Device fails if clamp voltage is not reached.				
End point electrical measurements		See table I, subgroup 2 herein, I_{CEX1} and h_{FE2} electrical tests				
Subgroups 6 and 7						
Not applicable						

1/ For sampling plan see MIL-S 19500.
2/ L = 5 mH (2 each Essex Stancor C-2688 in parallel 1A, 0.5 Ω, or equivalent (see 4.5.2)).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:42Z
Check the source to verify that this is the current version before use.



TURN-ON (t_{on}) TIME TEST CIRCUIT



TURN-OFF (t_{off}) TIME TEST CIRCUIT

FIGURE 2. Switching time test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:42Z
Check the source to verify that this is the current version before use.

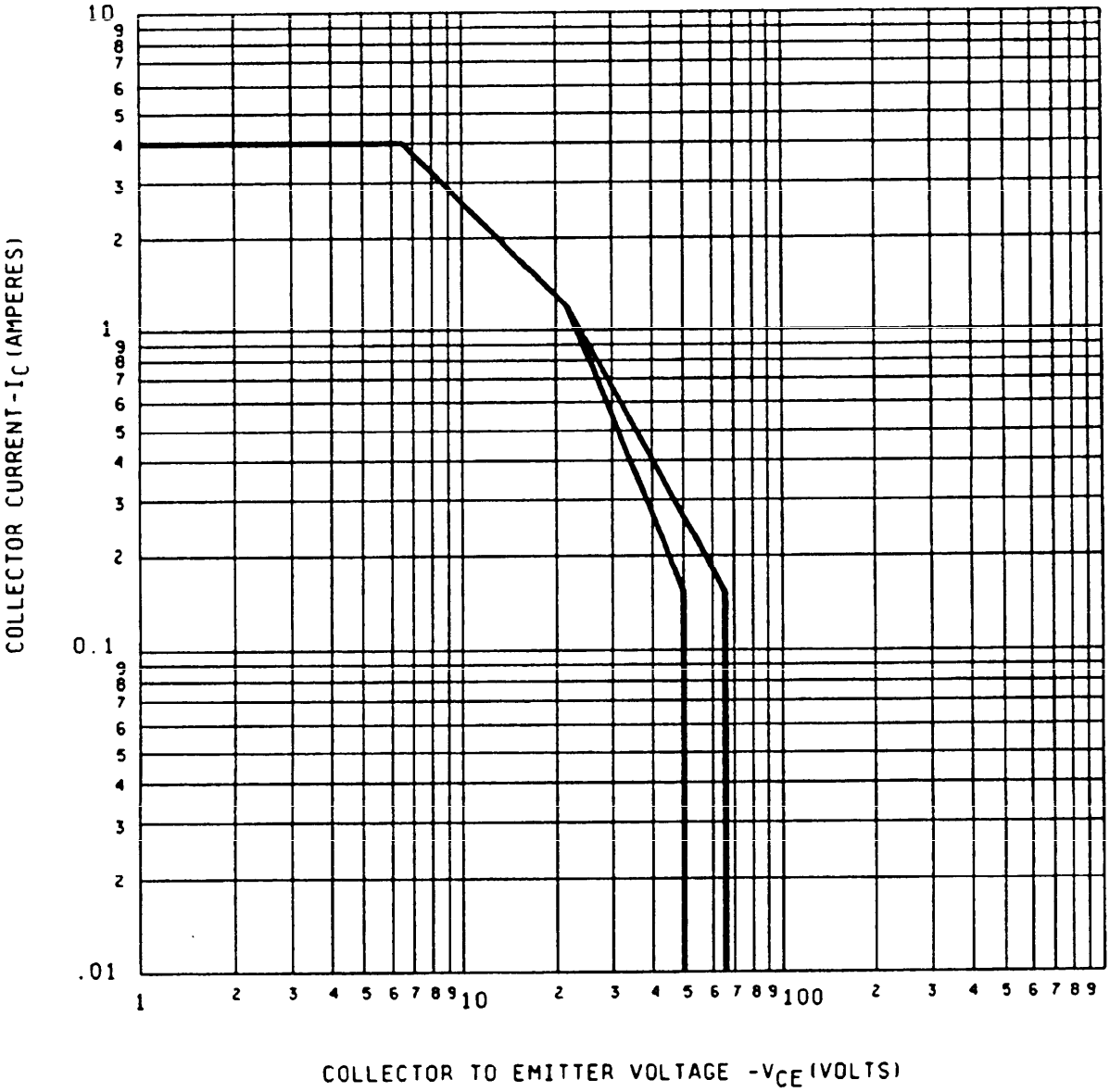


FIGURE 3. Maximum SOA graph (continuous dc) for types 2N3766 and 2N3767.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:42Z
Check the source to verify that this is the current version before use.

4.5.3 Thermal resistance. Thermal resistance measurements shall be conducted in accordance with method 3131 of MIL-STD-750. The following details shall apply:

a. Collector current magnitude during power applications shall be 0.4 A dc.

b. Collector to emitter voltage magnitude shall be 20 V dc.

c. Reference temperature measuring point shall be the case.

d. Reference point temperature shall be +25°C ≤ T_R ≤ +35°C and recorded before the test is started.

e. Mounting arrangement shall be with heat sink to case.

f. Maximum limit shall be $R_{\theta JC}$ = 7°C/W.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation

b. Lead finish as specified (see 3.3.1).

c. Type designation and product assurance level.

6.3 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Air Force - 17

Review activities:
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
DLA - ES

(Project 5961-F100)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T02:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/518B(USAF)	2. DOCUMENT DATE (YYMMDD) 94/11/21

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, POWER, TYPE 2N3766, 2N3767, JAN, JANTX, AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048	(2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) efense Electronics Supply Center ATTN: DESC-ELDT Dayton, OH 45444-5765	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T02:42Z
Check the source to verify that this is the currentversion before use.